

**NANYAMBA TOWN COUNCIL
NITEKELA SECONDARY SCHOOL
FORM TWO SATURDAY EXAMINATION SERIES #2
CODE : 032 CHEMISTRY**

TIME:2:30 HRS

FEBRUARY, 2026

INSTRUCTIONS

1. This paper consists of three sections A, B and C with total of **ten** Questions.
2. Section A and C carries **15** Marks and section B carries **70** marks.
3. All writings must be **Blue** or **Black ink** except for diagrams which must be in **pencil**.

SECTION A (15 Marks)

Answer All Questions

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1. For each of the items (i) – (x), choose the correct answer from among the given alternatives and write its letter in the box provided. **(10 Marks)**

(i) Which of the following is a compound?

- A. Helium (He) B. Hydrogen (H₂) C. Carbon dioxide (CO₂) D. Neon (Ne)

(ii) Your brother was given a syrup with a prescription that IT SHOULD BE WELL SHAKEN BEFORE USE. This is because the syrup is a

- A. Clear solution B. Suspension C. Homogenous mixture D. Filtrate

(iii) The components of a mixture of iron and sand can be separated by

- A. Boiling B. Chemical reaction C. Magnetic attraction D. Filtration

(iv) Your chemistry teacher denied a student to enter the laboratory because of a floppy clothing. This is because the cloth may

- A. Be blown by wind B. Affect the measurements
C. Catch fire or cause one to fall D. Cause poor ventilation in the body

(v) Imagine that, you are playing a chemistry related trivia game with your friends. If you are to choose one term which does not relate to any recognized branches of chemistry among the following, which term could you choose?

- A. Organic chemistry B. Arithmetic chemistry C. Inorganic chemistry D. Physical chemistry

(vi) If a form two aspires to be a professional farmer, which among the following subjects a student should be advised to focus on in order to develop the necessary competences?

- A. Geography, Physics and Chemistry B. Biology, Physics and Mathematics
C. Agriculture, Chemistry and Biology D. Chemistry, Physics and Biology

(vii) Why non – luminous flame is the most applicable flame for heating and welding purposes:

- A. It is very noise B. It is very hot C. It has no soot D. It has three zones

(viii) The following apparatus is used to keep test tubes in the laboratory

- A. Test tube holder B. Test tube rack C. Test tube brush D. Basin for test tubes

(ix) Mr. Mwaisa coated the roofs of his house with zinc in order to protect it from rusting. What was the process done by Mr. Mwaisa?

- A. Painting B. Galvanization C. Oiling D. Alloying

(x) When a chemist studies a substance, he/she is interested in its

- A. Shape B. Properties C. Smell D. Force of attraction

i	ii	iii	iv	v	vi	vii	viii	ix	x

2. Match the descriptions in LIST A with the correct techniques for separating mixtures in LIST B by writing the letter of the correct response in the table provided. **(05 Marks)**

LIST A	LIST B
(i) Extraction of 5 g of table salt dissolved in 10 cm ³ of salt solution.	A. Funnel separation
(ii) Separation of cooking oil from a mixture of water and cooking oil.	B. Solvent extraction
(iii) Collecting 50 cm ³ of pure water from 300 L copper (II) sulphate solution.	C. Evaporation
(iv) Extraction of flower pigment from hibiscus flower using 2.5 cm ³ of ethanol.	D. Fractional distillation
(v) Separating serum from suspension of human blood	E. Centrifugation
	F. Simple distillation
	G. Paper chromatography

i	ii	iii	iv	v

SECTION B (70 Marks)

Answer All Questions

3. (a) Some of elements have their symbols completely different from their common English names.

(i) Where are their symbols derived from? **(01 mark)**

(ii) Use the table below to list three elements of this kind **(03 Marks)**

ENGLISH NAME	LATIN NAME

(b) The teacher told Chamtemakuni to list two elements which he knows. Chamtemakuni wrote Al and O. Why did he use letters instead of writing their full names? Give three (3) reasons **(03 Marks)**

(c) Briefly give reasons for each of the following laboratory rules **(03 Marks)**

(i) Do not return unused chemicals in their original container, why?

(ii) Open the windows while you are working in laboratory, why?

(iii) Laboratory doors and windows must open outwards, why?

4. (a) The figure below is a part of the modern periodic table which includes the first 20 elements **(08 Marks)**

Q		D					C
H	M					L	

(i) Mention the chemical symbols of the elements represented by letters

Q_____

(ii) Write the electronic configuration of the elements represented by the Letters

M

C

(iii) Which letter represent element that belong to group III?

(iv) Which letters represent elements that belongs to the same group?

(v) Which letters represent elements that belongs to the same period?

(b) Using clear statement distinguish between Mendeleev's periodic law from Modern periodic law **(02 Marks)**

5. (a) Draw the structure of an atom according to the: (*label it if possible*) (04 Marks)

Dalton's Atomic Theory	Rutherford's Atomic Model

(b) Many scientists explained about the structure of an atom including John Dalton. State four postulates of Dalton's Atomic Theory. **(04 Mark)**

(c) State two components of Nucleons.

(02 Marks)

6. (a) State the number of protons, neutrons and electrons in each of the following atoms.

(06 Marks)

(i) $^{27}_{13}\text{Al}$ (ii) $^{235}_{92}\text{U}$

$^{27}_{13}\text{Al}$	$^{235}_{92}\text{U}$
Protons =	Protons =
Neutrons =	Neutrons =
Electrons =	Electrons =

(b) Mention two application of isotopes

(04 Marks)

7. (a) A Librarian of Nitekela secondary school found that there is a poor light supply in the library room, as a result the quality of the room was low. He decided to find the causes of the problem by observing the main electrical switch. Unfortunately, the fire erupted and spread in the room whereas a large number of books and magazines burnt due to fire.

(07 Marks)

(i) State the class of fire that caused the burning of large number of books and magazines. Give reason a for your choice

(ii) What type of a potable fire extinguisher should have been used to extinguish the fire so as to save some books and magazines? Give a reason for your choice.

(iii) What type of a potable fire extinguisher should have been not used to extinguish the fire? Give a reason.

(iv) Before extinguishing the fire using an appropriate fire extinguisher, what action should have been taken first? Explain

(b) Why some chemistry apparatus are made of porcelain/clay materials? Give three reasons

(03 Marks)

8. (a) How does chemistry contribute to the development of today's modern society. Give five points **(05 Marks)**

(b) Since chemistry is the science that involves experiments and practical works, chemists have to acquire scientific skills in order to be successful in obtaining facts and verifying them. Mention Five (5) skills to be acquired by chemists. **(05 Marks)**

9. (a) Ten students from Nitekela secondary school were doing an experiment in the school laboratory with all windows closed in such a way that low circulation of air occurred in the room. Some containers of chemicals such as concentrated acid were giving out fumes hence made the students to suffocate. If you were among the students in the school laboratory, what are two (2) things that you would have done in order to rescue yourself and others from suffering the same problem as those occurred to the students? **(04 Marks)**

(b) The study of Chemistry provides chemists with the knowledge and skills that are required in their future career opportunities. Mention any six professions that chemists can acquire in the future. **(03 Marks)**

(c) A good chemistry laboratory has various qualities and characteristics. As a form two student from Nitekela secondary school Mention any three qualities of a good chemistry laboratory you know. **(03 Marks)**

SECTION C (15 Marks)

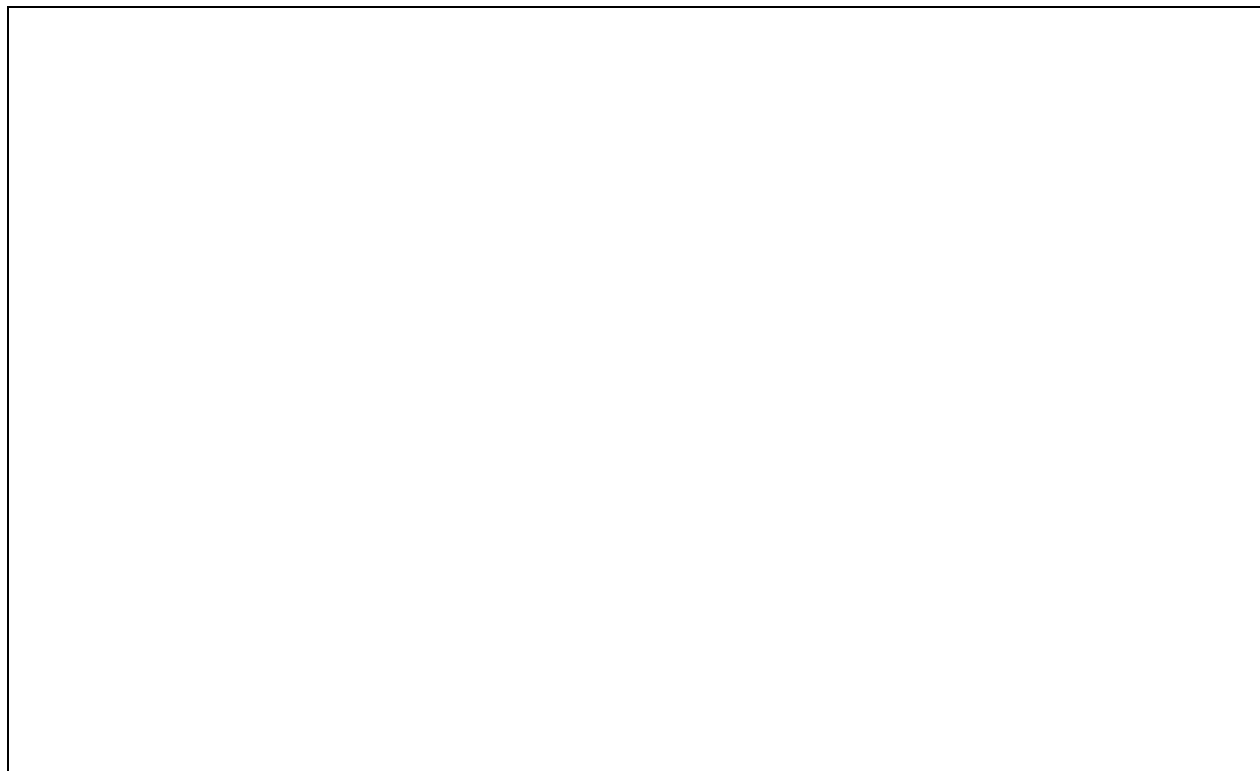
Answer Question 10

10. (a) Give four differences between the flame formed when air holes are open and when air holes are closed

(06 Marks)

Flame when air holes closed	Flame when air holes opened

(b) Draw a well labeled diagram of a small laboratory heat source consisting of a vertical metal tube connected to a gas source (04 Marks)



(b) Describe how you will light a device drawn above (05 Marks)

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“Curiosity is the Greatest Asset in Chemistry, Don’t be Afraid to Explore”
Sir Medrack Mponzi